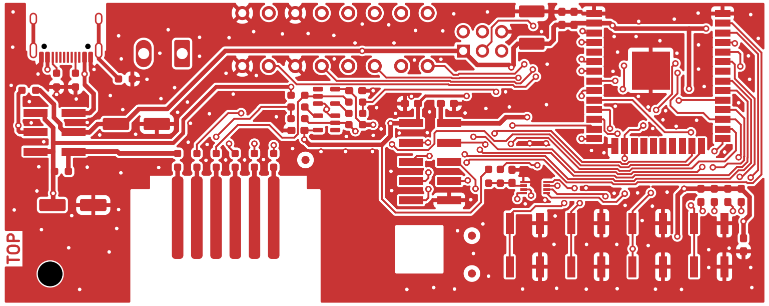
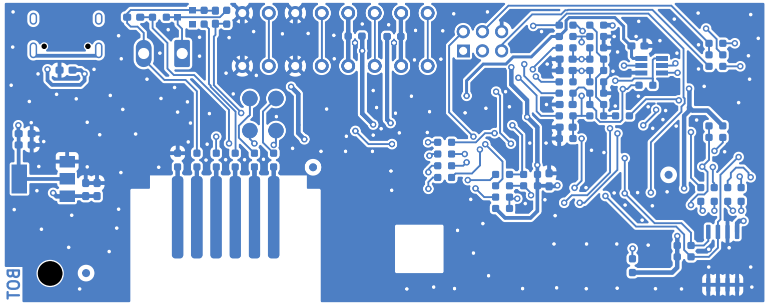
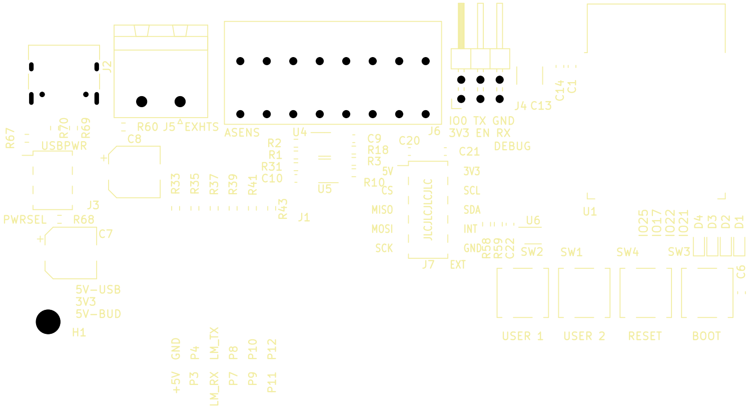
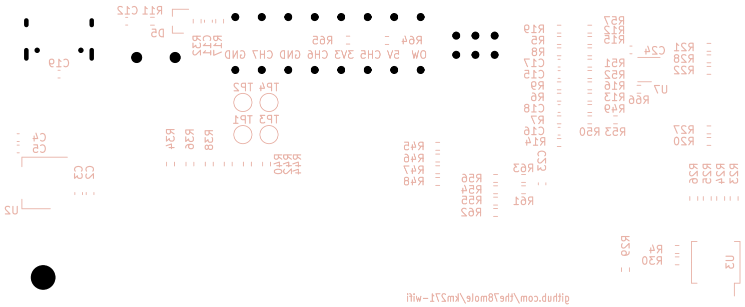


TOP

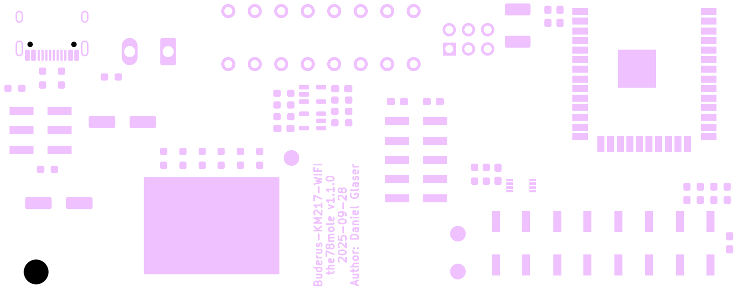

BOT

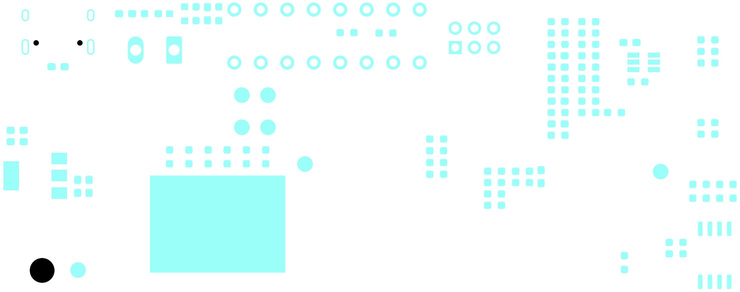


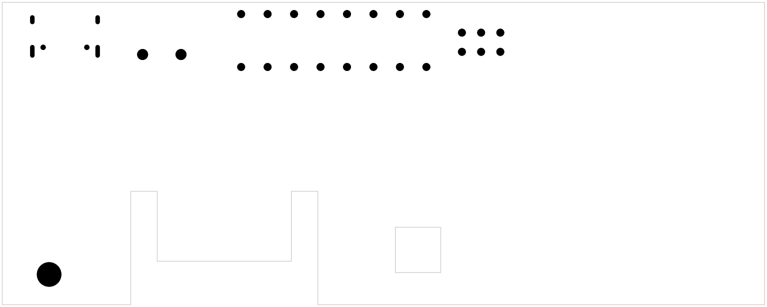
J2
R67
R70
R69
USBPWR
R60
J5
EXHTS
C8
ASENS
U4
R2
R1
R31
C10
U5
C9
R18
R3
R10
C20
J6
3V3
IO0
TX
EN
GND
RX
DEBUG
J4
C13
C14
C1
C21
5V
CS
MISO
MOSI
SCK
SCL
SDA
INT
GND
JLCJLCJLCJLC
J7
EXT
R58
R59
C22
U6
U1
IO25
IO17
IO22
IO21
D4
D3
D2
D1
SW2
SW1
SW4
SW3
C6
USER 1
USER 2
RESET
BOOT
R33
R35
R37
R39
R41
R43
J1
J3
PWRSEL
R68
C7
5V-USB
3V3
5V-BUO
H1
+5V
GND
P3
P4
LM_RX
LM_TX
P7
P8
P9
P10
P11
P12

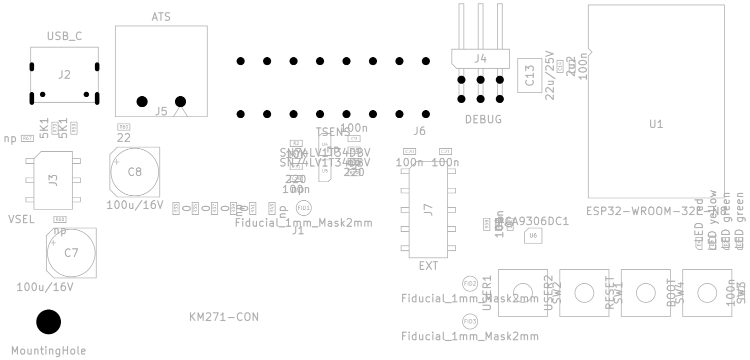


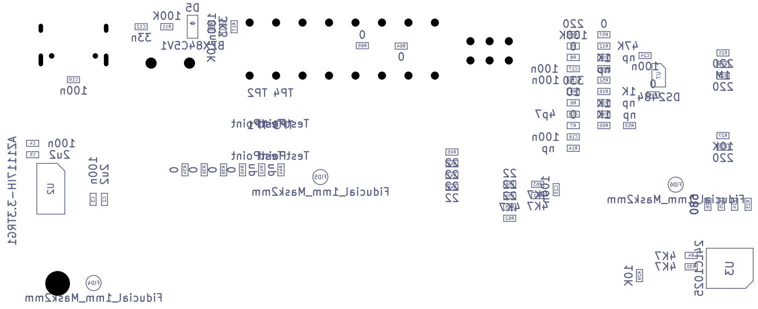
0W 5V CH5 3V3 CH6 GND CH7 GND
TP1
TP2
TP3
TP4
U2
U3
U7
D5
github.com/the78mole/km271-wifi

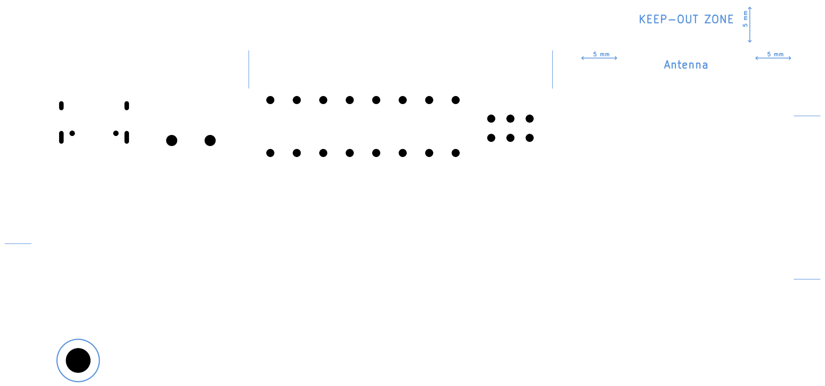

Buderus-KM217-WIFI
the78mole v1.1.0
2025-09-28
Author: Daniel Glaser






USB_C
J2
ATS
J5
J4
C13
22u/25V
2u2
100n
U1
ESP32-WROOM-32E-N8
DEBUG
J6
TSENS
100n
SN74LV1T34DBV
SN74LV1T34DBV
220
220
100n
100n
100n
J7
EXT
np
5K1
5K1
22
J3
VSEL
np
C8
100u/16V
C7
100u/16V
np
Fiducial_1mm_Mask2mm
J1
PCA9306DC1
100n
USER1
USER2
SW2
RESET
SW1
BOOT
SW4
100n
SW3
LED red
LED yellow
LED green
LED green
Fiducial_1mm_Mask2mm
Fiducial_1mm_Mask2mm
KM271-CON
MountingHole


D5
100K
33n
BZX84C5V1
TP4 TP2
TestPoint
TestPoint
0
100n
100n
2u2
AZ1117IH-3.3TRG1
U2
Fiducial_1mm_Mask2mm
Fiducial_1mm_Mask2mm
Fiducial_1mm_Mask2mm
220
100K
47K
100n
DS2484
1K
4p7
1M
10K
220
22
4K7
U3
10K



FAB NOTE J1 (KM271 card edge): 12 gold fingers F+B, 1.6 mm FR4 (+-10%).
Order option: Gold Fingers = YES (ENIG / hard gold on fingers), mask open over fingers.
Bevel: NOT POSSIBLE - finger edge (y=124.25) is recessed 5.75 mm between the board legs
(y=130) and JLC chamfers only boards >= 50x50 mm. Original module is used unbevelled.
Fingers end 0.2 mm before the board edge; no vias/pads within 1 mm (rule area GOLD_FINGERS).

FAB NOTES Buderus-KM217-WIFI rev v1.1.0
Stackup: 2 layers, FR4 TG135, 1.6 mm +-10 pct, outer copper 1 oz (35 um)
L1 F.Cu signal/GND pour | core 1.5 mm | L2 B.Cu signal/GND pour
Finish: ENIG (gold fingers). Gold fingers: YES, bevel: NO (see J1 note)
Mask green, silk white, vias tented. Min track/space 0.20/0.20 mm, min via 0.3/0.6 mm
Assembly: Standard PCBA, both sides. BOM/CPL: production/bom.csv, production/positions.csv